Model Behavior

February 2026

1 Project Overview

In this project, we aim to predict the final transaction price (ClosePrice) of residential properties in California using machine learning Models, including

- Linear Regression (Baseline),
- Decision Tree,
- Decision Tree (new features),
- Random Forest,
- Random Forest (new features),
- XGBoost (tuned),
- LightGBM (tuned).

The input features consist of physical attributes (e.g., Bed Bath Ratio, Lot Utilization), location-based factors (e.g., Zip Median Price), and temporal features (e.g., Months From Dec 2025). The target variable is LogClosePrice, which reduces extreme skewness and improves the stability of the models.

2 Evaluation Metrics

We evaluate model performance using four complementary metrics:

- R^2 (on actual price): Measures how well the model explains variance in true house prices, reflecting overall goodness of fit in the original scale.
- R^2 (on log(ClosePrice)): Assesses model performance in the transformed space, which is directly aligned with the training objective and less sensitive to extreme values.
- MAPE (Mean Absolute Percentage Error): Captures the average relative prediction error, making it easy to interpret model accuracy in percentage terms.
- MdAPE (Median Absolute Percentage Error): Reports the median of percentage errors, providing a robust measure of typical prediction accuracy.

3 Model Performances Comparison

We tested the following 7 models

- Linear Regression – basic model, straightforward relationship between features and price. Here we use only PostalCode as the predictor, which is the feature with the highest single-feature R^2.
- Decision Tree (improved) – A tree-like model captures non-linear relationships and handles feature interactions automatically. In many models we encode location using the median log sale price per ZIP code calculated from the training data only. This provides a strong numeric signal for neighborhood price levels while avoiding high dimensionality.
- Decision Tree (new features) – same tree model but using additional engineered features. In this model we use Core Property Attributes: LivingArea, BedroomsTotal, BathroomsTotalInteger; Location Information: ZipMedianPrice; Engineered Features: BedBathRatio, SqftPerBedroom, LotUtilization; and Time and Age Factors: PropertyAge MonthsFromDec2025. We omit the feature PPSF (Price per Square Foot) because it causes data leakage.
- Random Forest – ensemble of decision trees to improve performance and reduce overfitting. It achieves high predictive accuracy and captures nonlinear relationships.
- Random Forest (new features) – The Random Forest model with additional engineered features achieves a very high R^2 score and very low MAPE and MdAPE. However, because it includes both PPSF and LivingArea—features that directly determine the target—this introduces data leakage, making the model's performance unrealistic.
- XGBoost – gradient boosting method, good for tabular data. Here again, we use ZipMedianPrice: obtained by taking the median of each postal code and assigning it to each row of the dataset by postal code. This XGBoost model is effective at producing accurate predictions of residential property closing prices, as indicated by the low MAPE and MdAPE values.
- LightGBM – another gradient boosting framework, optimized for speed and accuracy. The LightGBM regression model was trained on log-transformed property prices using feature-engineered variables, including ZIP-level median prices, property attributes, and derived metrics such as SqftPerBedroom and LotUtilization.

We summarized the key characteristics and main features of the seven models as shown in figure 1:

	Model	Features	Target	Key Characteristics
0	Linear Regression	PostalCode (one-hot)	LogClosePrice	Simple linear model, only captures ZIP-level average effect, log-transform stabilizes variance
1	Decision Tree	ZipMedianPrice, LivingArea, BathroomsTotalInteger, BedroomsTotal	LogClosePrice	Tree-based model, captures nonlinear interactions, uses ZIP-level median price encoding, max_depth=6 and min_samples_leaf=100
2	DT (new features)	LivingArea, BedroomsTotal, BathroomsTotalInteger, ZipMedianPrice, Bed_Bath_Ratio, Sqft_Per_Bedroom, Lot_Utilization, Property_Age, Months_From_Dec_2025	LogClosePrice	ZIP Median Price Feature, max_depth=6, min_samples_leaf=100
3	Random Forest	All features (one-hot)	LogClosePrice	Handles nonlinearities; n_estimators=500; max_features='sqrt'
4	RF (new features)	LivingArea, BathroomsTotalInteger, BedroomsTotal, PPSF, Bed_Bath_Ratio, Property_Age, Months_From_Dec_2025, Sqft_Per_Bedroom, Lot_Utilization.	LogClosePrice	Data leakage because PPSF * LivingArea ≈ Price
5	XGBoost (tuned)	LivingArea, BathroomsTotalIntege, BedroomsTotal, Bed_Bath_Ratio, Property_Age, Months_From_Dec_2025, Sqft_Per_Bedroom, Lot_Utilization, ZipMedianPrice	LogClosePrice	ZIP Median Price Feature, Best Parameters: 'subsample': 0.7, 'n_estimators': 500, 'min_child_weight': 5, 'max_depth': 5, 'learning_rate': 0.1, 'gamma': 0, 'colsample_bytree': 0.7
6	LightGBM (tuned)	ZipMedianPrice, LivingArea, BathroomsTotalInteger, BedroomsTotal, Bed_Bath_Ratio, Property_Age, Months_From_Dec_2025, Sqft_Per_Bedroom, Lot_Utilization	LogClosePrice	Use feature-engineered variables, including ZIP-level median prices, Best hyperparameters: 'subsample': 0.9, 'reg_lambda': 1, 'reg_alpha': 5, 'num_leaves': 150, 'n_estimators': 200, 'min_child_samples': 10, 'max_depth': -1, 'learning_rate': 0.05, 'colsample_bytree': 0.8}

Figure 1: Summary of Models and Key Characteristics

We use R^2 both log and original scale and MAPE, MdAPE to evaluate the tested models. The following figure 2 is the Model Performance Comparison.

	Model	R^2	R^2 (log)	MAPE (%)	MdAPE (%)
0	Linear Regression	0.590000	0.726900	23.900000	17.080000
1	Decision Tree	0.784200	0.847300	17.770000	12.520000
2	Decision Tree (new features)	0.699100	0.849900	17.910000	13.130000
3	Random Forest	0.812100	0.865900	15.980000	10.430000
4	Random Forest (new features)	0.989600	0.998500	0.600000	0.300000
5	XGBoost	0.839700	0.896200	14.440000	10.190000
6	LightGBM	0.837600	0.892500	14.580000	10.120000

Figure 2: Model Evaluations Comparison

The performance of Random Forest model with new features is almost too perfect. In real-world house price prediction, such near-zero errors strongly suggest: Overfitting, or Data leakage. Actually, PPSF contributes about 59% and LivingArea contributes about 41% of the model's importance. This means the model is essentially using price (Price Per Square Foot) and LivingArea to predict price, which is a form of data leakage. Because $PPSF \times LivingArea \approx Price$, the two variables together almost reconstruct the target variable directly.

XGBoost is the Best Practical Model: $R^2 = 0.8397$ (highest among realistic models), R^2 (log) $= 0.8962$ (best), MAPE $= 14.44\%$, MdAPE $= 10.19\%$ (very strong). This model

achieves the best balance across all metrics, strong generalization, and handles nonlinear relationships well.

LightGBM is Very close to XGBoost: Slightly lower R^2 and R^2 (log) Slightly higher error. This model is the second best, but difference is very small.

4 Forward Holdout Results

Show how your model performed on January 2026 data. (Preparing)